

COMPUTER NETWORKS PROJECT
ZEROth REVIEW
QoS OPTIMISATION FOR VIDEO STREAMING

GROUP : D-15

PROBLEM STATEMENT :

Video streaming over networks suffers from quality degradation due to:

- Network congestion - Videos buffer and stop playing
- Variable bandwidth - Quality keeps changing (HD to SD to blurry)
- Packet loss - freezing
- High latency - Delays in video playback
- Inefficient resource utilization - Network capacity is wasted

OBJECTIVES :

- Implement Intelligent Buffering System
 - Build adaptive buffer management that changes size based on network conditions
 - Implement predictive algorithms that anticipate network problems
- Create Dynamic Quality Adaptation
 - Automatically switch video quality (HD/SD/Low) based on available bandwidth
 - Use proactive rather than reactive quality adjustment
- Develop Multi-Queue Packet Scheduling
 - Implement priority queues
 - Create smart algorithms to decide which queue gets each packet
- Build Network Condition Simulator
 - Simulate real-world network problems (congestion, packet loss, variable bandwidth)
 - Test system performance under different conditions
- Measure and Compare Performance
 - Compare my optimized system against basic streaming
 - Measure improvements in buffering time, quality consistency, and user experience

IMPLEMENTATION :

- TCP/UDP socket programming for client-server communication
 - Network performance measurement tools (latency, throughput, jitter)
 - Basic file streaming with chunked data transfer
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- Video file handler that breaks videos into small chunks
 - Client-side buffer that stores incoming video chunks
 - Network condition simulator that adds realistic network problems
 - Real-time monitoring system for network performance

QoS OPTIMISATION ALGORITHMS :

1. Intelligent Buffering Algorithm
2. Dynamic Quality Adaptation Algorithm
3. Hybrid TCP/UDP Protocol

QUALITY METRICS :

- **Packet Loss Ratio (PLR)**

What it measures: Percentage of video packets that fail to reach destination

Formula:

$$PLR = (\text{Packets_Sent} - \text{Packets_Received}) / \text{Packets_Sent} \times 100\%$$

Units: Percentage (%)

- 0-0.5%: Excellent
- 0.5-1%: Good
- 1-3%: Fair
- 3%: Poor (noticeable artifacts)

- **End-to-End Delay (Latency)**

What it measures: Time between packet transmission and reception

Components:

- Transmission delay
- Propagation delay
- Queuing delay
- Processing delay

Units: Milliseconds (ms)

- <50ms: Real-time applications (excellent)
- 50-150ms: Interactive applications (good)

- 150-400ms: Streaming applications (acceptable)
- 400ms: Poor user experience

○ **Throughput**

What it measures: Actual data transfer rate achieved

Formula:

$$\text{Throughput} = \text{Total_Bytes_Received} / \text{Time_Duration}$$

Units:

- bps (bits per second)
- Kbps, Mbps, Gbps
 - 4K Video: 15-25 Mbps
 - 1080p HD: 3-8 Mbps
 - 720p: 1.5-4 Mbps
 - 480p: 0.5-2 Mbps

○ **Quality Switches**

What it measures: How often video quality changes during playback

Types:

- **Upward switches:** Low → High quality
- **Downward switches:** High → Low quality
- **Switch frequency:** Switches per minute

Units: Switches per minute

- <1/min: Good adaptation
- 1-2/min: Acceptable
- 3/min: Too frequent (jarring)